

EDUCATIONAL QUALITY ANALYSIS THROUGH STUDENT FEEDBACK

*A Project Report submitted in partial fulfillment of the
requirements for the award of degree of*

**BACHELOR OF TECHNOLOGY
in
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING**

Submitted

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**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
ADITYA UNIVERSITY**

(Formerly Aditya Engineering College (A))

2025 - 2028

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



ADITYA
UNIVERSITY

CERTIFICATE

This is to certify that the project work entitled “**EDUCATIONAL QUALITY ANALYSIS THROUGH STUDENT FEEDBACK**” is being submitted by

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DECLARATION

We hereby declare that the project entitled “**EDUCATIONAL QUALITY ANALYSIS THROUGH STUDENT FEEDBACK**” is a genuine project. This work has been submitted to the **ADITYA UNIVERSITY (Formerly Aditya Engineering College (A))**, Surampalem in partial fulfillment of the **B.Tech.** degree. We further declare that this project work has not been submitted in full or part of the award of any degree or any other educational institutions.

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Not to forget, Parents **Faculty, Teaching, Non-Teaching Staff and our Friends** who have directly or indirectly supported us in completing this project on time.

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ABSTRACT

Student feedback is an essential component in evaluating and improving the quality of teaching and the overall educational experience. Educational institutions collect large volumes of feedback through surveys and rating-based questionnaires; however, manual analysis of such data is time-consuming, subjective, and inefficient. To address this issue, this project applies data mining techniques to analyze student feedback systematically using computational tools.

The objective of this project is to identify student satisfaction levels and discover meaningful patterns from feedback data. The dataset consists of student responses collected as ratings for multiple questions related to teaching effectiveness, course content, classroom interaction and overall learning experience. Data preprocessing is performed using Python to ensure data quality, including handling missing values, removing inconsistencies, and transforming categorical data into numerical form with the help of Pandas and NumPy libraries.

Feature engineering techniques are applied to identify significant attributes influencing student satisfaction. Data mining methods such as classification techniques categorize students into satisfaction levels such as highly satisfied, satisfied, and dissatisfied. These techniques are implemented in the Google Colab environment.

Data visualization using the Matplotlib library presents results through graphs and charts for better understanding of trends.

The results provide valuable insights into student perceptions, enabling educational institutions to evaluate teaching effectiveness, identify areas for improvement, and make data-driven decisions to enhance academic quality and student satisfaction.

KEYWORDS :

Student Feedback Analysis, Data Mining, Dataset Collection, Data Preprocessing, Classification, Student Satisfaction, Educational Data Mining, Data Visualization.

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EDUCATIONAL QUALITY ANALYSIS THROUGH STUDENT FEEDBACK

CHAPTER 1 : INTRODUCTION

1.1 Introduction of the Project Domain

The project titled Educational Quality Analysis Through Student Feedback using Data Mining focuses on analyzing student feedback to evaluate the quality of education.

Educational institutions collect large amounts of feedback from students regarding teaching effectiveness, course content, classroom interaction, and overall learning experience.

However, analyzing this feedback manually is difficult, time-consuming, and may lead to inaccurate conclusions. This application uses data mining techniques and computational tools such as Python and WEKA to process and analyze feedback data efficiently.

The system helps in identifying student satisfaction levels and extracting meaningful patterns from feedback data. This enables institutions to improve teaching quality and make better academic decisions.

1.2 Existing System and Disadvantages

Existing System

In the current system, student feedback is collected through paper forms or simple online forms and stored in registers or spreadsheets. The analysis of this feedback is done manually by teachers or administrators. They calculate averages and review responses one by one to understand student satisfaction. There is no proper automated system to process or analyze large amounts of feedback data. As a result, the process becomes slow and less efficient.

Disadvantages

The existing system has some drawbacks:

- Manual errors may occur during calculation and analysis
- Time-consuming process when handling large datasets
- Difficult to identify patterns and trends in feedback

- Data may be inconsistent or incomplete
- Lack of accurate classification of student satisfaction
- Limited visualization and reporting capabilities

1.3 Proposed System and Advantages

The proposed system is an automated Educational Quality Analysis Through Student Feedback developed using Python and data mining techniques. It collects student feedback data and processes it efficiently using computational tools. The system performs data preprocessing to clean the dataset by handling missing values, removing duplicates, and correcting inconsistencies. It then applies classification algorithms such as Decision Tree and Naïve Bayes using WEKA to analyze the data and categorize students into satisfaction levels like Highly Satisfied, Satisfied, and Dissatisfied. The results are presented using graphs and charts, making them easy to understand. This system reduces manual effort, saves time, and improves accuracy by minimizing human errors. It can handle large amounts of data efficiently and helps in identifying patterns and trends in feedback. Overall, the system is reliable, user-friendly, and supports better decision-making to improve teaching quality and student experience.

1.4 Objectives of the Project

The main objective of this project is to analyze student feedback using data mining techniques in an efficient and accurate way. The system aims to preprocess and clean the dataset by handling missing values and removing inconsistencies. It also focuses on identifying important factors that affect student satisfaction. Another objective is to classify students into categories such as Highly Satisfied, Satisfied, and Dissatisfied using machine learning algorithms. The project also aims to present the results using graphs and charts for better understanding. Overall, it helps educational institutions make better decisions to improve teaching quality and student experience.

1.5 Organization of the Project

This project is organized into different chapters to explain each part clearly. Chapter 1 provides the introduction, problem statement, objectives, and scope of the project. Chapter 2 explains the system overview, including the existing system, its limitations, and the proposed system. Chapter 3 describes the technologies used such as Python, WEKA, and data mining techniques. Chapter 4 focuses on the system design and modules involved in the project. Chapter 5 explains the implementation and working of the system. Finally, the last chapter presents the conclusion and future enhancements of the project.

CHAPTER 2 : REQUIRMENT ANALYSIS

2.1 Introduction

Requirement analysis is an important phase in the development of a system, where the needs and expectations of users are clearly identified. In this project, the requirement analysis focuses on understanding how student feedback can be collected, processed, and analyzed effectively. It helps in identifying the type of data required, such as feedback ratings and student details, and the operations to be performed on that data. This phase also defines the functional and non-functional requirements of the system. By analyzing these requirements, the system can be designed to work efficiently, provide accurate results, and meet the needs of educational institutions.

2.2 Hardware & Software Requirements

Hardware Requirements

The minimum hardware required for this system is:

- Processor: Intel i3 or above
- RAM: 4 GB (8 GB recommended for better performance)
- Hard Disk: At least 10 GB free space
- Monitor, Keyboard, and Mouse

Software Requirements

The software required for developing and running this project includes:

- Python (for data preprocessing and analysis)
- WEKA (for data mining and classification)
- Google Colab / Jupyter Notebook
- Libraries: Pandas, NumPy, Matplotlib, Scikit-learn
- Operating System: Windows / Linux / macOS

2.3 Software & Hardware Requirements Specification

The software and hardware requirements specification provides a detailed explanation of the components required for the system to function efficiently. The system requires a computer with a minimum Intel i3 processor and at least 4 GB RAM, while 8 GB RAM is recommended for better performance and smooth execution of data analysis tasks. A minimum of 10 GB free disk space is necessary to store datasets, program files, and output results.

On the software side, Python is used for data preprocessing, analysis, and implementation of data mining techniques. Libraries such as Pandas and NumPy help in handling and processing large datasets efficiently, while Matplotlib is used for visualization. Scikit-learn supports machine learning operations. WEKA is used for applying classification algorithms and analyzing patterns in the data. Tools like Jupyter Notebook or Google Colab provide an environment for executing the code.

These requirements are necessary to ensure that the system runs smoothly, processes data accurately, and delivers reliable results without performance issues.

CHAPTER 3 : LITERATURE SURVEY

3.1 Related Works

Several research works have been carried out in the field of student feedback analysis and educational data mining. Many researchers have used data mining and machine learning techniques to analyze student feedback and improve educational quality. Classification algorithms such as Decision Tree, Naïve Bayes, and Random Forest are commonly used to predict student satisfaction levels.

These existing works show that automated feedback analysis systems are more efficient and accurate compared to manual methods. They also help institutions make better decisions to improve teaching effectiveness and student satisfaction.

3.2 Survey Table

The survey table presents a summary of various research works related to student feedback analysis using different techniques. From the table, it can be observed that several methods such as Decision Tree, Naïve Bayes, and Random Forest are commonly used for classification and prediction of student satisfaction. These methods help in improving the accuracy and efficiency of analysis. Some research works also focus on data preprocessing techniques to clean and prepare the dataset, which enhances data quality. Additionally, visualization techniques are used to represent the results in the form of graphs, making it easier to understand patterns and trends. Overall, the survey shows that combining data preprocessing, classification algorithms, and visualization leads to better analysis of student feedback.

Student Name	Roll No	Department	Section	Q1_Teaching effectiveness	Q2_Content Quality	Q3_Classroom Interaction	Q4_Resources	Q5_Overall Experience
Harsha Naik	R1000	EEE	C	5	2	4	1	1
Aditya Sharma	R1002	CSE	C	4	4	4	3	3
Tajja Rao	R1003	CIVIL	A	4	5	5	5	2
Rohan Kumar	R1004	EEE	C	1	2	2	4	5
Harsha Rao	R1005	IT	C	1	1	4	5	2
Sanika Rao	R1006	EEE	B	5	1	1	3	1
Ananya Naik	R1007	IT	C	1	1	1	1	1
Parvati Rao	R1008	CIVIL	A	3	3	1	5	4
Divya Kumar	R1009	EEE	B	2	2	2	2	1
Ravathi Rao	R1010	CIVIL	A	2	5	1	1	4
Nha Sharma	R1011	EEE	B	4	5	1	5	3
Ananya Rao	R1012	EEE	B	1	3	3	5	5
Ravathi Naik	R1013	IT	A	5	4	1	1	2
Rahul Reddy	R1014	EEE	C	1	2	1	5	3
Kanya Naidu	R1015	MEEH	B	3	3	4	5	5
Tajja Kumar	R1016	EEE	C	5	2	3	5	4
Ravathi Naik	R1017	IT	B	2	5	1	1	2
Tajja Kumar	R1018	EEE	A	4	5	4	4	3
Harsha Naik	R1019	EEE	C	2	5	4	5	2
Vishnu Kumar	R1020	CIVIL	C	4	2	5	3	2
Arun Naidu	R1021	CSE	C	2	2	1	2	5
Vishnu Rao	R1022	IT	C	4	2	1	2	1
Ananya Rao	R1023	IT	B	3	1	1	2	1
Rahul Kumar	R1024	EEE	B	4	2	1	4	3
Parvati Kumar	R1025	CSE	B	5	2	2	4	1
Parvati Kumar	R1026	EEE	C	4	4	4	3	5
Nikhil Sharma	R1027	EEE	B	2	4	3	4	5
Ananya Naik	R1028	EEE	B	4	5	1	4	1
Kanya Naidu	R1029	IT	C	1	1	2	5	1
Divya Sharma	R1030	CIVIL	B	4	5	5	3	5

Survey Table

CHAPTER 4 : MODULES

The system is divided into several modules to perform different tasks efficiently.

4.1 Modules

The system is divided into different modules to make the design simple, organized, and easy to understand. Each module performs a specific task and works together with other modules to complete the overall process. This modular approach helps in better management, easy development, and maintenance of the system.

In this project, the modules handle different stages such as data collection, preprocessing, analysis, and visualization. By dividing the system into modules, the complexity is reduced and each part can be developed and tested independently. This improves the efficiency and performance of the system.

4.2 Module Implementation

4.2.1 Data Collection Module

This module is responsible for collecting student feedback data through surveys or forms. It gathers information such as teaching effectiveness, course content, classroom interaction, and overall experience. The collected data is stored for further processing. It ensures that the input data is properly organized.

4.2.2 Data Preprocessing Module

This module cleans and prepares the dataset for analysis. It handles missing values, removes duplicate records, and corrects inconsistent data. It also converts categorical data into numerical format. Preprocessing improves the quality and accuracy of the data.

4.2.3 Data Mining / Classification Module

This module applies machine learning techniques to analyze the data. It uses classification algorithms like Decision Tree and Naïve Bayes. The system categorizes students into satisfaction levels such as Highly Satisfied, Satisfied, and Dissatisfied. It helps in discovering meaningful patterns.

4.2.4 Data Visualization Module

This module presents the analysis results in graphical form. It uses charts such as bar graphs and pie charts. Visualization makes it easier to understand trends and patterns. It helps users interpret the results clearly.

CHAPTER 5 : SYSTEM DESIGN

5.1 Introduction

System design is an important phase in the development of a project where the overall structure and working of the system are planned. It defines how different components of the system interact with each other to achieve the desired functionality. This phase helps in converting the requirements into a structured design.

In this project, system design explains the architecture of the student feedback analysis system, including data flow, modules, and processing steps. It shows how data is collected, processed, analyzed, and presented as output. Proper system design ensures that the system is efficient, reliable, and easy to understand and implement.

5.2 Process Flow of the Project

The process flow of the project describes the step-by-step working of the student feedback analysis system. It begins with the collection of student feedback data through surveys or forms. The collected data is then loaded into the system for processing.

Next, data preprocessing is performed to clean the dataset by removing duplicates, handling missing values, and correcting inconsistent data. After preprocessing, feature engineering is applied to calculate important values such as the average feedback score.

The processed data is then analyzed using data mining techniques. Classification algorithms such as Decision Tree and Naïve Bayes are used to categorize students into satisfaction levels like Highly Satisfied, Satisfied, and Dissatisfied.

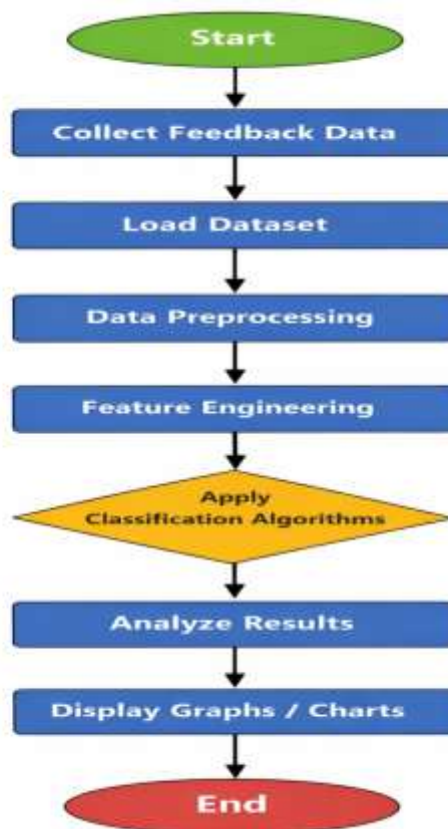
Finally, the results are displayed using graphs and charts for easy understanding. The output helps in identifying patterns and improving teaching quality based on student feedback.

5.3 Flowchart Diagram

The flowchart diagram represents the step-by-step process of the student feedback analysis system in a visual form. It shows how data flows from input to output through different stages of the system.

The process starts with the collection of student feedback data. The collected data is then given as input to the system. In the next step, data preprocessing is performed, which includes handling missing values, removing duplicates, and correcting inconsistent data. After preprocessing, feature engineering is carried out to calculate important values such as the average feedback score.

The processed data is then passed to the data mining stage, where classification algorithms like Decision Tree and Naïve Bayes are applied to analyze the data and categorize student satisfaction levels. Finally, the results are displayed in the form of graphs and charts. The process ends after presenting the output.



6. SYSTEM IMPLEMENTATION

6.1 INTRODUCTION

System implementation is the phase where the designed system is developed and executed using appropriate tools and technologies. It involves converting the system design into a working model by writing code, integrating modules, and testing the overall functionality. This phase ensures that all components of the system work together correctly.

In this project, the implementation is carried out using Python and data mining tools. The system processes student feedback data through various stages such as data collection, preprocessing, analysis, and visualization. Each module is implemented step by step to ensure accuracy and efficiency. Proper implementation helps in achieving the project objectives and delivering reliable results.

6.2 Selection Software

The selection of software is an important step in developing the project, as it determines the efficiency and performance of the system. In this project, Python is selected as the main programming language because of its simplicity and strong support for data analysis and machine learning. WEKA is chosen as a data mining tool due to its user-friendly interface and ability to apply various classification algorithms easily.

Libraries such as Pandas and NumPy are used for data preprocessing and manipulation, while Matplotlib is used for visualization of results. Tools like Jupyter Notebook or Google Colab are selected for executing the code in an interactive environment. These software tools are chosen because they are easy to use, efficient, and suitable for handling large datasets and performing accurate analysis.

6.3 Steps Involved in Implementation of Algorithm/Technology

The implementation of the student feedback analysis system is carried out in a step-by-step process to ensure accurate results. First, the dataset is collected from student feedback surveys containing ratings for different parameters such as teaching quality and course content. The collected data is then loaded into the system using Python.

In the next step, data preprocessing is performed to clean the dataset. This includes handling missing values, removing duplicate records, and correcting inconsistent data. After cleaning, feature engineering is applied to calculate important values such as the average feedback

score for each student.

Once the data is prepared, classification algorithms such as Decision Tree and Naïve Bayes are applied using WEKA. These algorithms analyze the data and classify students into satisfaction levels like Highly Satisfied, Satisfied, and Dissatisfied.

Finally, the results are visualized using graphs and charts with the help of Matplotlib. This step helps in understanding patterns and trends in the data. The complete process ensures that the system provides accurate and meaningful insights from student feedback.

CHAPTER 7 : TESTING

7.1 Introduction

Testing is an important phase in software development used to ensure that the system works correctly and meets the required objectives. It helps in identifying errors, bugs, and issues in the system before final deployment. In this project, testing is performed to verify that the student feedback analysis system processes data accurately and produces correct results. Different types of testing are carried out to check data preprocessing, classification, and visualization modules. The system is tested with sample datasets to ensure that it handles missing values, removes duplicates, and correctly classifies student satisfaction levels. Proper testing ensures the reliability, accuracy, and performance of the system.

7.2 Test Cases

The system uses a dataset consisting of student feedback collected through surveys, which includes ratings for teaching effectiveness, course content, classroom interaction, and overall experience on a scale of 1 to 5. After cleaning and processing, the data is used to calculate average scores and classify students into satisfaction levels such as Highly Satisfied, Satisfied, and Dissatisfied.

Test Cases					
Test Case ID	Module	Test Description	Input	Expected Output	Status
TC-01	Data Loading	Verify dataset loading	Student feedback dataset	Dataset loads successfully	Pass ✓
TC-02	Preprocessing	Check missing value handling	Dataset with missing values	Missing values handled	Pass ✓
TC-03	Preprocessing	Remove duplicate records	Dataset with duplicates	Duplicates removed	Pass ✓
TC-04	Data Cleaning	Validate data consistency	Raw dataset	Clean dataset generated	Pass ✓
TC-05	Feature Engineering	Calculate average score	Feedback ratings	Correct average calculated	Pass ✓
TC-06	Classification	Apply algorithm	Processed dataset	Correct classification achieved	Pass ✓
TC-07	Visualization	Generate graphs/charts	Analyzed data	Graphs and charts displayed	Pass ✓
TC-08	Output Display	Show final results	Classified data	Results displayed to user	Pass ✓

CHAPTER 8 : SCREENS & REPORTS

8.1 Introduction

This chapter presents the screens and reports of the student feedback analysis system. Screens represent the user interface through which users interact with the system, such as data input, processing, and result display. These screens are designed to be simple and user-friendly for easy operation.

Reports show the output generated by the system after analyzing the data. They include results such as student satisfaction levels, processed data, and graphical representations like charts and graphs. These reports help in understanding the analysis clearly and support better decision-making.

8.2 Screenshots

The output screens display the results generated by the student feedback analysis system. These screens present the processed data in a clear and understandable format. After applying data preprocessing and classification techniques, the system shows the final results such as student satisfaction levels, average scores, and categorized data.

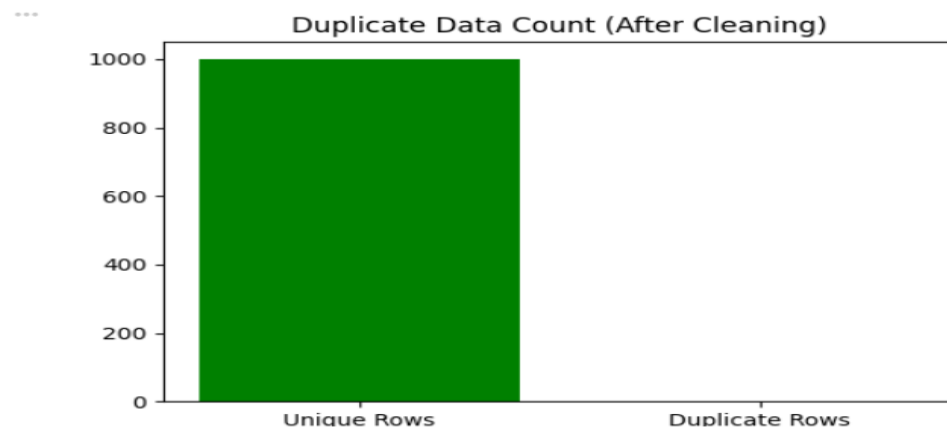
The output is also represented using graphical formats like bar charts and pie charts, which help in visualizing patterns and trends in the feedback data. These screens are designed to be user-friendly and easy to interpret. They help users quickly understand the analysis results and support effective decision-making for improving educational quality.

8.2.1 Data Preprocessing

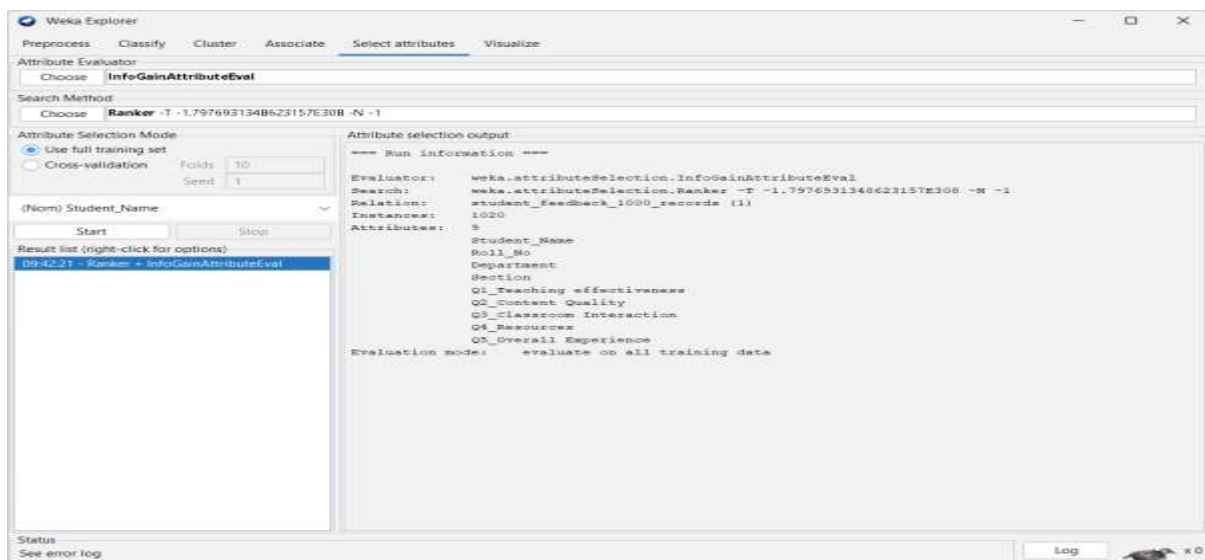
Handling Missing Values:

```
# After Cleaning
duplicates_after = df_clean[df_clean.duplicated()]
plt.figure(figsize=(6,4))
plt.bar(["Unique Rows", "Duplicate Rows"],
        [df_clean.shape[0]-duplicates_after.shape[0],
         duplicates_after.shape[0]],
        color=["green","red"])

plt.title("Duplicate Data Count (After Cleaning)")
plt.show()
```



Attribute Slection Using Ranker:



After Preprocessing:

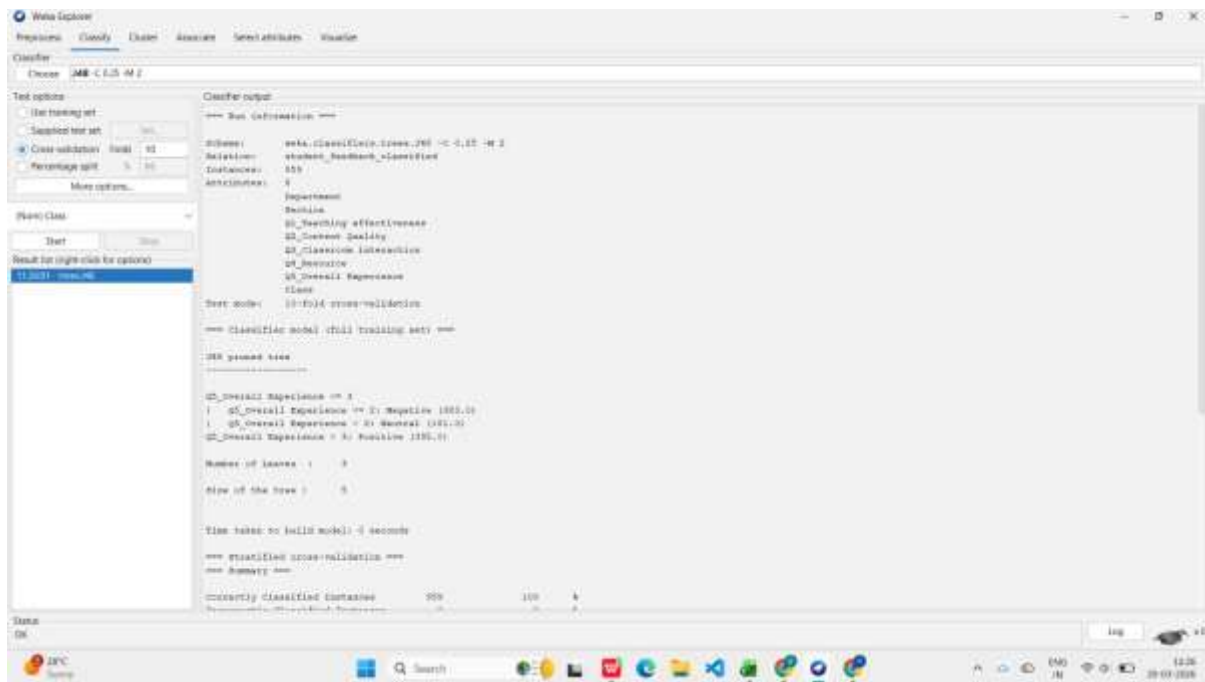
ID	Name	Roll No.	Program	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
1000	Aditya Sharma	10000	IT	B	5	3	3	4	1	4			
1001	Aditya Sharma	10001	MECH	B	3	5	4	4	4	5			
1002	Aditya Sharma	10002	IT	A	4	2	3	1	5				
1003	Aditya Sharma	10003	CSE	E	1	1	3	3	3	4			
1004	Aditya Sharma	10004	CIVIL	A	4	3	4	5	2				
1005	Aditya Sharma	10005	MECH	B	1	2	5	4	2	5			
1006	Aditya Sharma	10006	CSE	B	4	1	5	2	2	4			
1007	Aditya Sharma	10007	MECH	C	2	1	1	2	4				
1008	Aditya Sharma	10008	CSE	A	4	2	3	3	4				
1009	Aditya Sharma	10009	CSE	C	5	3	4	4	3				
1010	Aditya Sharma	10010	IT	C	2	3	2	2	3				
1011	Aditya Sharma	10011	CIVIL	A	3	5	3	2	2				
1012	Aditya Sharma	10012	IT	C	4	1	4	3	3				
1013	Aditya Sharma	10013	CSE	A	4	5	1	4	5				
1014	Aditya Sharma	10014	CSE	A	5	5	4	1	4				
1015	Aditya Sharma	10015	CSE	B	3	2	2	2	3				
1016	Aditya Sharma	10016	CSE	B	5	3	5	5	5				
1017	Aditya Sharma	10017	IT	C	5	1	1	2	2				
1018	Aditya Sharma	10018	IT	C	5	2	4	2	1				
1019	Aditya Sharma	10019	CIVIL	B	2	4	4	2	1				
1020	Aditya Sharma	10020	IT	B	1	3	1	1	2				
1021	Aditya Sharma	10021	MECH	A	4	3	4	1	2				
1022	Aditya Sharma	10022	CIVIL	A	3	2	4	2	2				
1023	Aditya Sharma	10023	MECH	B	4	2	2	1	1				
1024	Aditya Sharma	10024	CSE	D	1	4	5	1	5				
1025	Aditya Sharma	10025	MECH	B	3	1	1	2	3				
1026	Aditya Sharma	10026	IT	A	2	4	5	2	5				
1027	Aditya Sharma	10027	CSE	A	1	5	3	1	2				
1028	Aditya Sharma	10028	CIVIL	E	5	5	2	2	4				
1029	Aditya Sharma	10029	CIVIL	C	2	5	5	2	5				

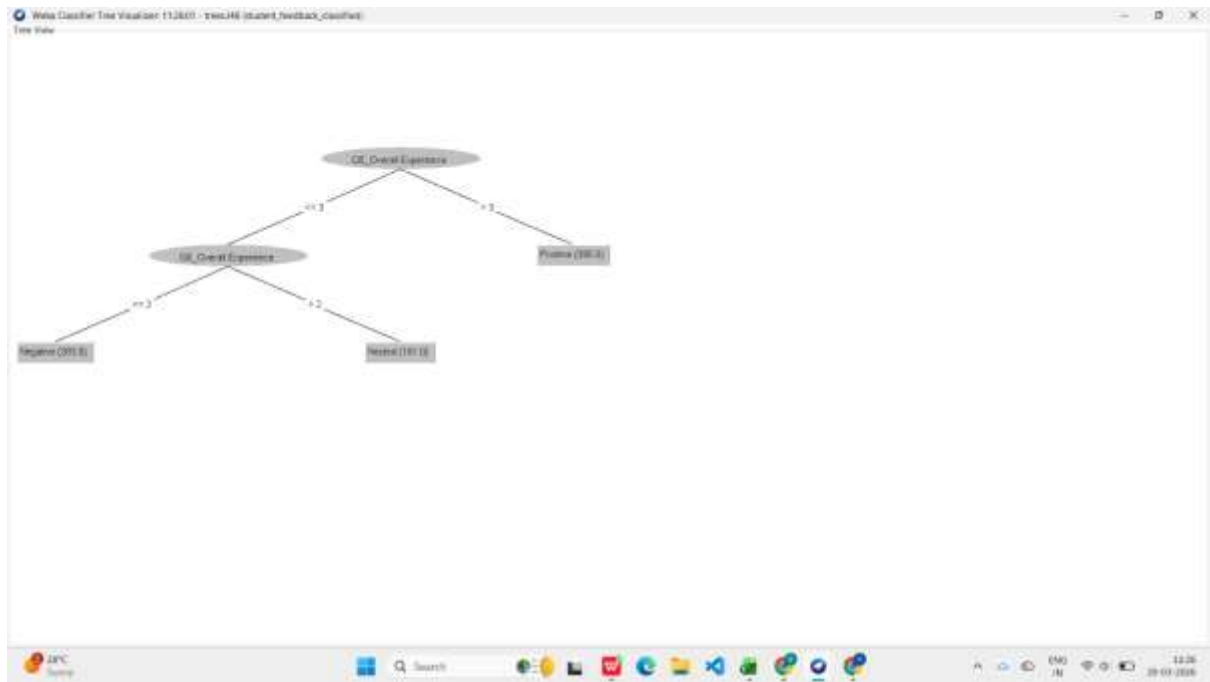
8.2.2 Data Classification

Data classification is a data mining technique used to categorize data into predefined groups based on certain conditions. In this project, data classification is used to analyze student feedback and group students into different satisfaction levels such as **Highly Satisfied, Satisfied, and Dissatisfied**.

After preprocessing the dataset, classification algorithms such as Decision Tree and Naïve Bayes are applied to the data. These algorithms analyze the feedback ratings and identify patterns to assign each record to a specific category. Classification helps in understanding student opinions clearly and makes it easier to evaluate teaching effectiveness.

The results of classification are further used to generate reports and visualizations, which provide meaningful insights and support decision-making for improving educational quality.





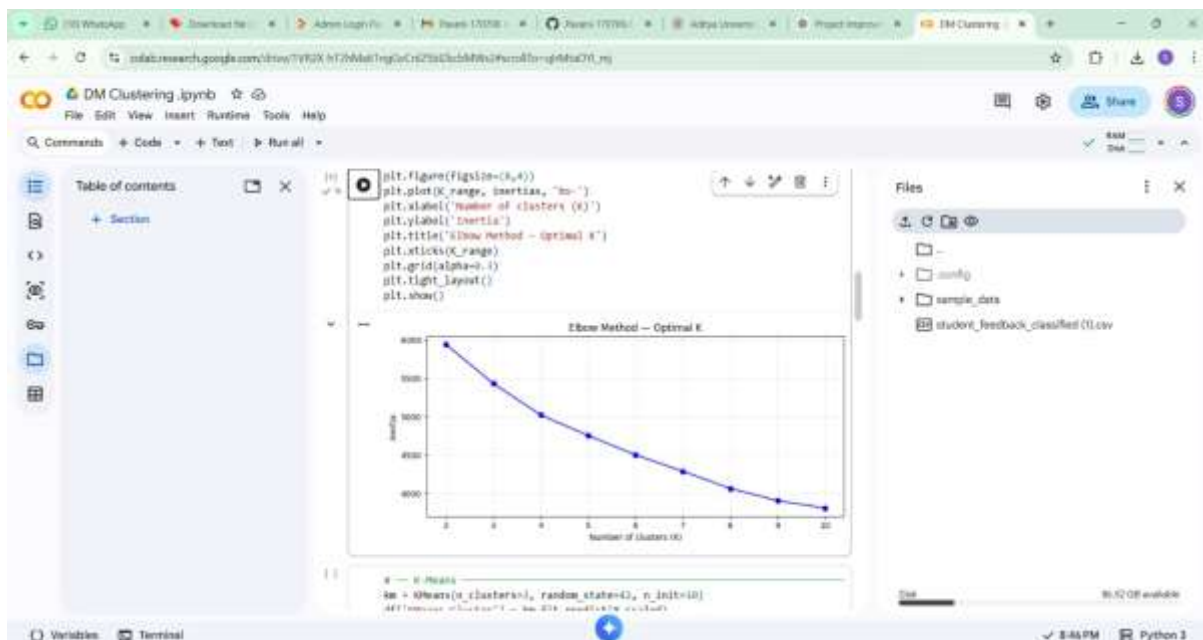
8.2.3 Data Clustering

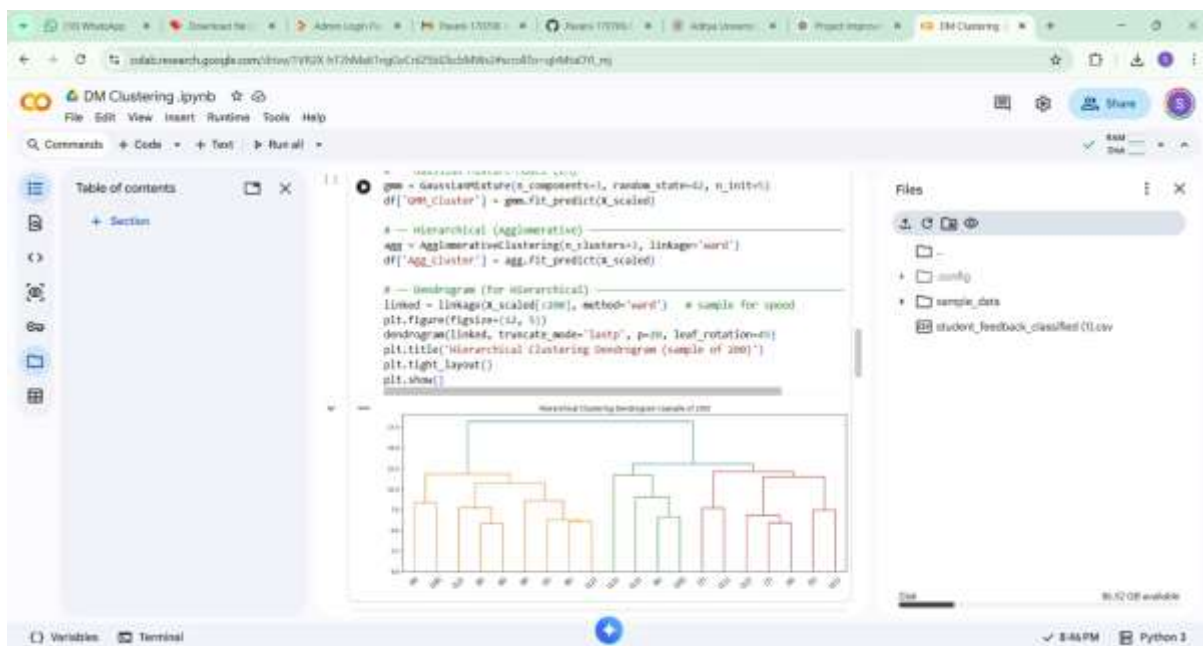
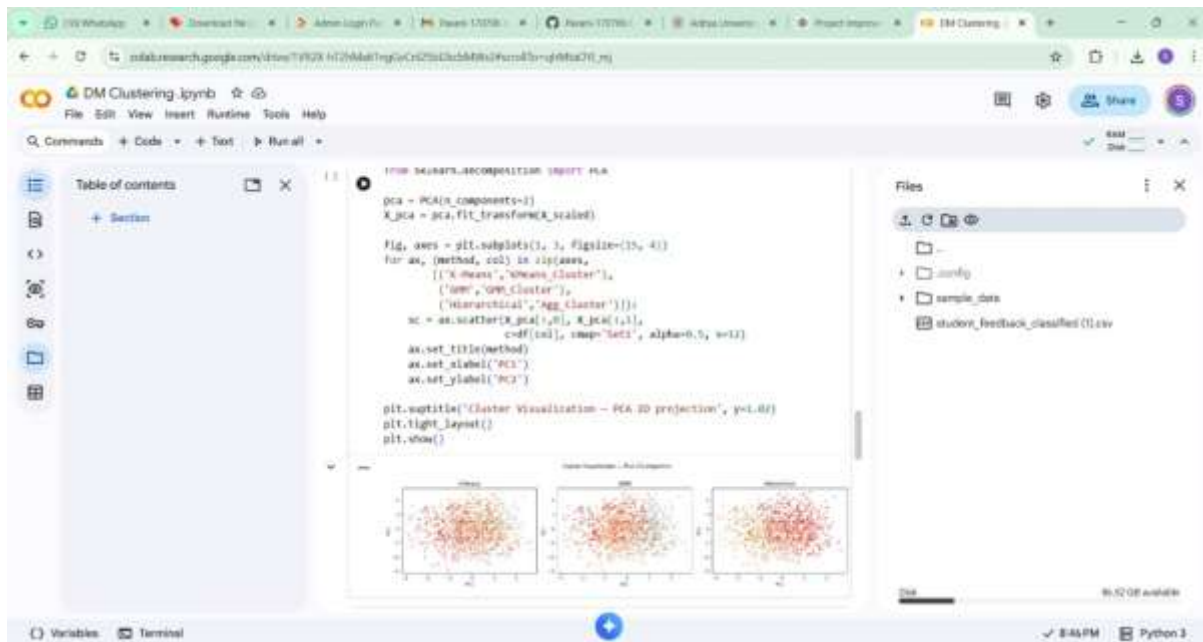
Data clustering is a data mining technique used to group similar data points together based on their characteristics without predefined categories. In this project, clustering is used to group students who have similar feedback patterns. For example, students with similar opinions on teaching quality, course content, or interaction are placed in the same group.

Clustering algorithms such as K-Means can be used to identify these groups automatically.

This helps in discovering hidden patterns and relationships in the data that are not easily visible through manual analysis. It also helps institutions understand different types of student behavior and feedback trends.

The results of clustering support better analysis and provide deeper insights into student satisfaction, which can be used to improve teaching methods and overall educational quality.

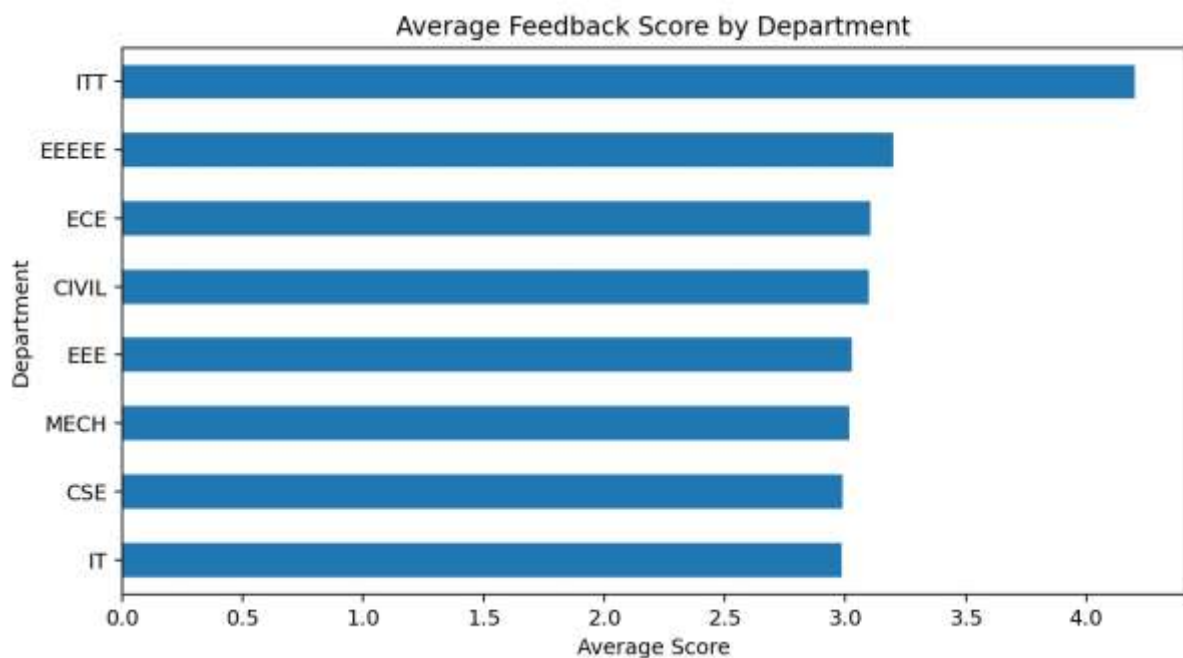


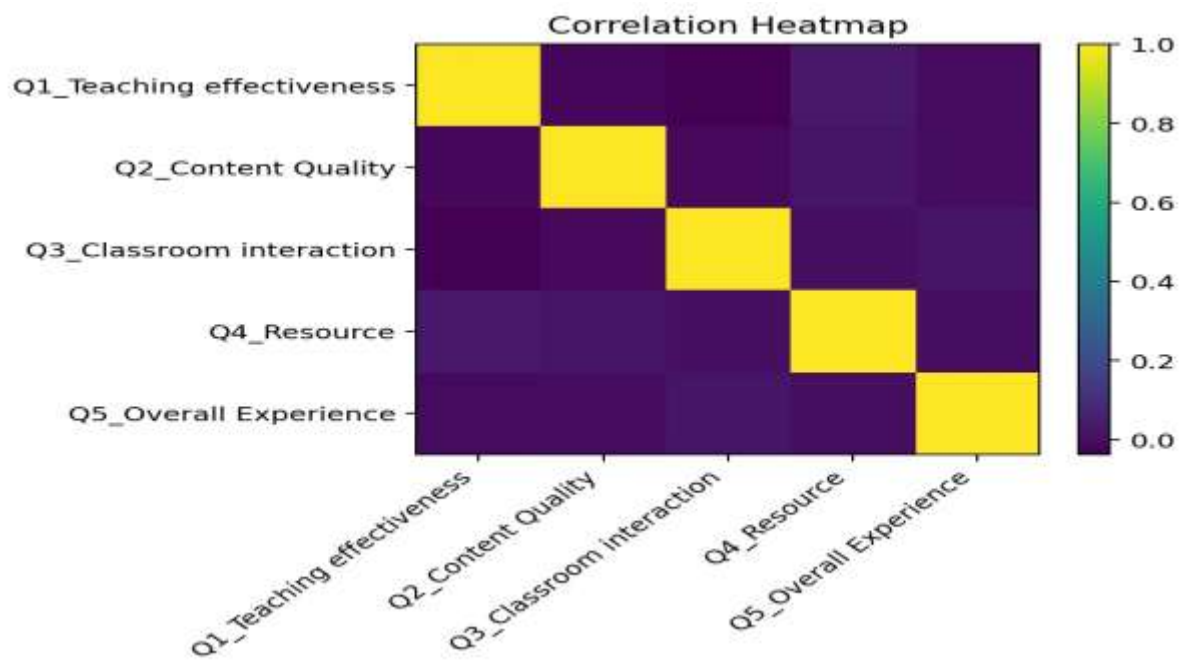


8.2.4 Data Visualization

Data visualization is the process of representing data in graphical form to make it easier to understand and analyze. In this project, data visualization is used to present the results of student feedback analysis in a clear and meaningful way. After processing and classification of the data, the results are displayed using charts such as bar graphs and pie charts.

These visualizations help in identifying patterns, trends, and comparisons in student feedback, such as satisfaction levels and performance of different parameters. The Matplotlib library in Python is used to generate these graphs. Data visualization makes the results more interactive and easy to interpret, helping educational institutions make better decisions to improve teaching quality and student experience.





Summary Statistics

Metric	Q1_Teaching effectiveness	Q2_Content Quality	Q3_Classroom interaction	Q4_Resource	Q5_Overall Experience
count	959.0	959.0	959.0	959.0	959.0
mean	3.06	3.07	3.05	3.0	3.02
std	1.45	1.4	1.35	1.38	1.43
min	1.0	1.0	1.0	1.0	1.0
25%	2.0	2.0	2.0	2.0	2.0
50%	3.0	3.0	3.0	3.0	3.0
75%	4.0	4.0	4.0	4.0	4.0
max	5.0	5.0	5.0	5.0	5.0

CHAPTER 9 : CONCLUSION

9.1 SUMMARY

The Educational Quality Analysis System Trough student Feedback helps in analyzing student feedback in an easy and efficient way. It reduces manual work and avoids errors that may occur during manual analysis. By using data mining techniques, the system can identify student satisfaction levels and important patterns in the data. The use of Python and WEKA improves accuracy and speed of analysis. It also helps in handling large amounts of data effectively. The visualization of results using graphs makes it easier to understand trends. Overall, the system helps educational institutions make better decisions to improve teaching quality and student experience. It also supports data-driven decision making and continuous improvement in education.

9.2 FUTURE SCOPE

The project can be further enhanced by integrating advanced machine learning algorithms to improve accuracy and prediction performance. A web-based or mobile application can be developed to collect feedback in real time. The system can be extended to include sentiment analysis for textual feedback. Integration with databases can help in managing large-scale data efficiently. Real-time dashboards and interactive visualizations can be added for better user experience. Additionally, the system can be implemented on a larger scale for universities and online learning platforms to improve educational quality.

CHAPTER 10 : REFERENCES

References for Project

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2. Tan, P.-N., Steinbach, M., & Kumar, V. (2019). *Introduction to Data Mining* (2nd Edition). Pearson.
3. WEKA Official Documentation:
<https://www.cs.waikato.ac.nz/ml/weka/documentation.html>
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8. Pandas Documentation: <https://pandas.pydata.org>
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11. Seaborn Documentation: <https://seaborn.pydata.org>
12. K-Means Clustering Tutorial: <https://scikit-learn.org/stable/modules/clustering.html#k-means>
13. Outlier Detection in Python: <https://towardsdatascience.com/outlier-detection-techniques-in-python-50f650c75aaf>
14. Student Feedback Form Analysis (Google Forms + Excel/Python):
<https://support.google.com/docs/answer/6281888>

CHAPTER 11 : APPENDIX

The appendix includes supporting materials and additional information used in the project, which provide deeper understanding and validation of the results obtained through Data Science technique

11.1 Source Codes

Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Load Dataset

```
df = pd.read_csv("student_attendance_dataset.csv")
df.head()
#Checking the dataset info
df.info()
A3. Classification with Data Handling
#Handle Missing Values
df["Attendance_Percentage"].fillna(df["Attendance_Percentage"].mean(), inplace=True)
```

#Handle Outliers #Check attendance

```
df.describe()
```

#Remove invalid attendance values

```
df = df[df["Attendance_Percentage"] <= 100]
df = df[df["Attendance_Percentage"] >= 0]
```

#Remove Duplicates

```
df = df.drop_duplicates()
```

#Feature Engineering (Calculate Total Marks)

```
df["Total_Marks"] = df["Assignment_Score"] + df["Midterm_Marks"] + df["Final_Marks"]
#Descriptive Statistics
df.describe()
```

#Histogram (Marks Distribution)

```
plt.hist(df["Total_Marks"], bins=10)
plt.xlabel("Total Marks")
plt.ylabel("Number of Students")
plt.title("Distribution of Student Marks")
plt.show()
```

#Box Plot (Attendance)

```
sns.boxplot(x=df["Attendance_Percentage"])
plt.title("Attendance Distribution")
plt.show()
```

#Scatter Plot (Attendance vs Marks)

```
plt.scatter(df["Attendance_Percentage"], df["Total_Marks"])
plt.xlabel("Attendance Percentage")
plt.ylabel("Total Marks")
plt.title("Attendance vs Marks")
plt.show()
```

#Heatmap

```
sns.heatmap(df.corr(numeric_only=True), annot=True) plt.show()
```

#Save Clean Dataset for WEKA

```
df.to_csv("clean_student_dataset.csv", index=False)
```

A4. Clusterization with Data Handling

#Elbow method – Optimal K

```
inertias = []
K_range = range(2, 11)
for k in K_range:
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(X_scaled)
    inertias.append(km.inertia_)
plt.figure(figsize=(8,4))
plt.plot(K_range, inertias, 'bo-')
plt.xlabel('Number of clusters (K)')
plt.ylabel('Inertia')
plt.title('Elbow Method — Optimal K')
plt.xticks(K_range)
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

#K-Means

```
km = KMeans(n_clusters=3, random_state=42, n_init=10)
df['KMeans_Cluster'] = km.fit_predict(X_scaled)
```

#GaussianMixtureModel

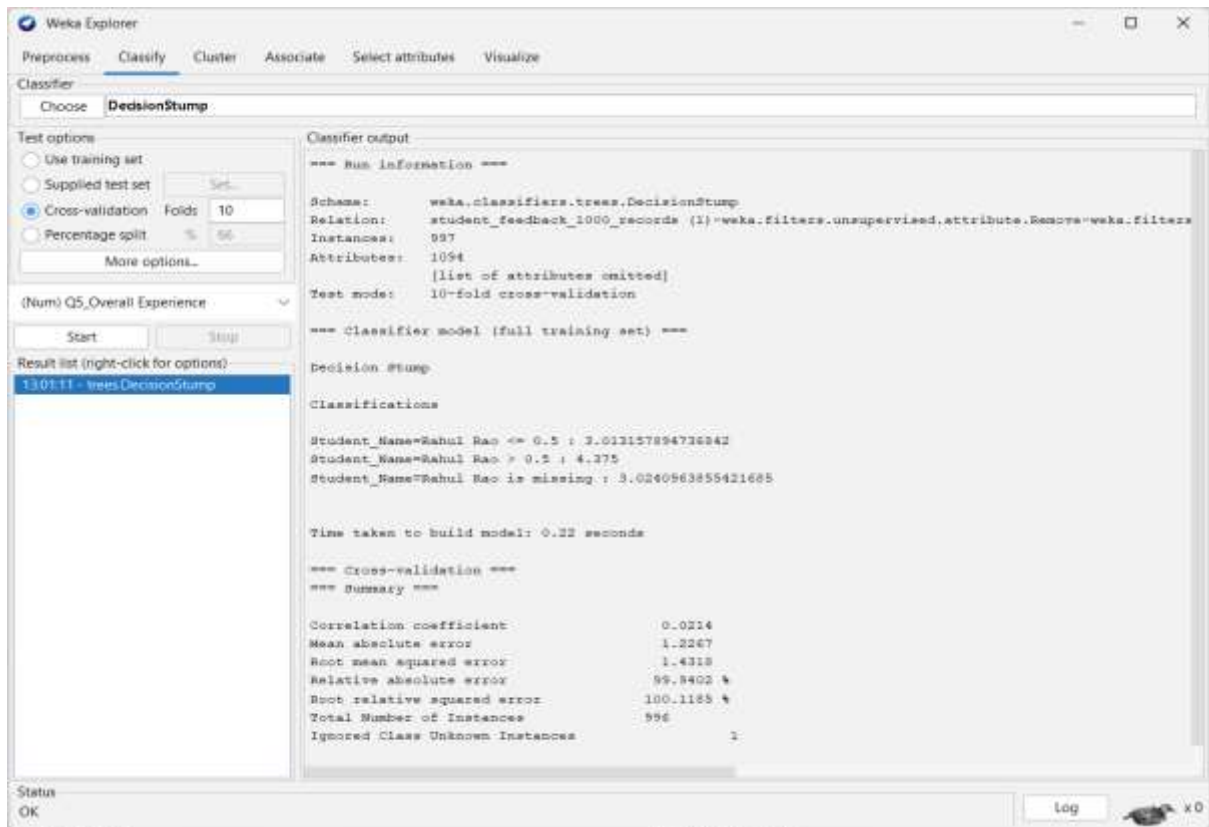
```
gmm = GaussianMixture(n_components=3, random_state=42, n_init=5)
df['GMM_Cluster'] = gmm.fit_predict(X_scaled)
```

#Hierarchical

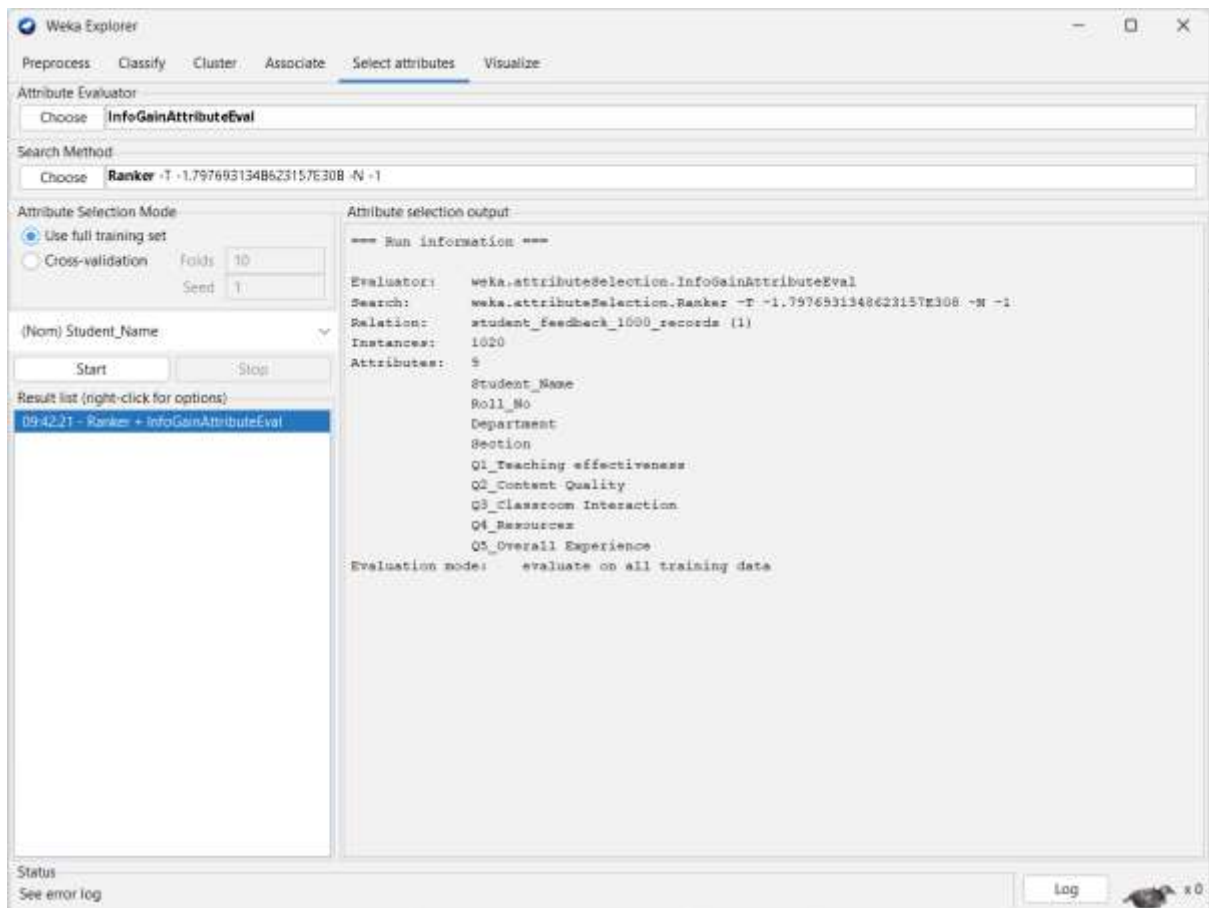
```
agg = AgglomerativeClustering(n_clusters=3, linkage='ward')
df['Agg_Cluster'] = agg.fit_predict(X_scaled)
```

#Dendrogram

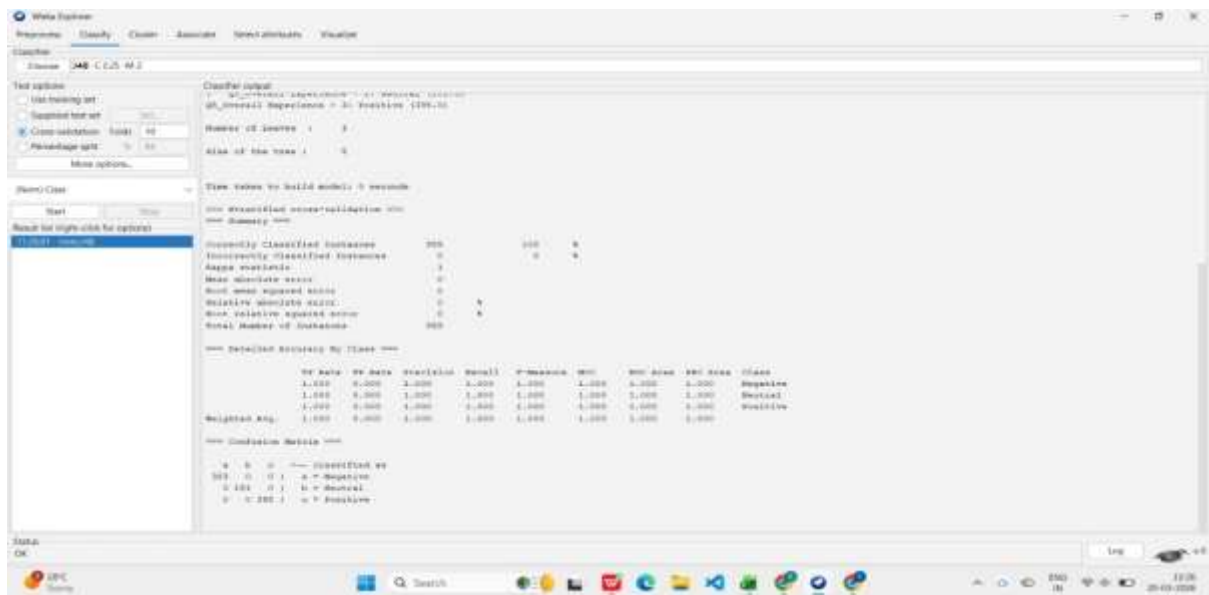
```
linked = linkage(X_scaled[:200], method='ward') # sample for speed
plt.figure(figsize=(12, 5))
dendrogram(linked, truncate_mode='lastp', p=20, leaf_rotation=45)
plt.title('Hierarchical Clustering Dendrogram (sample of 200)')
plt.tight_layout()
plt.show()
```



1.1 DecisionTreeStump



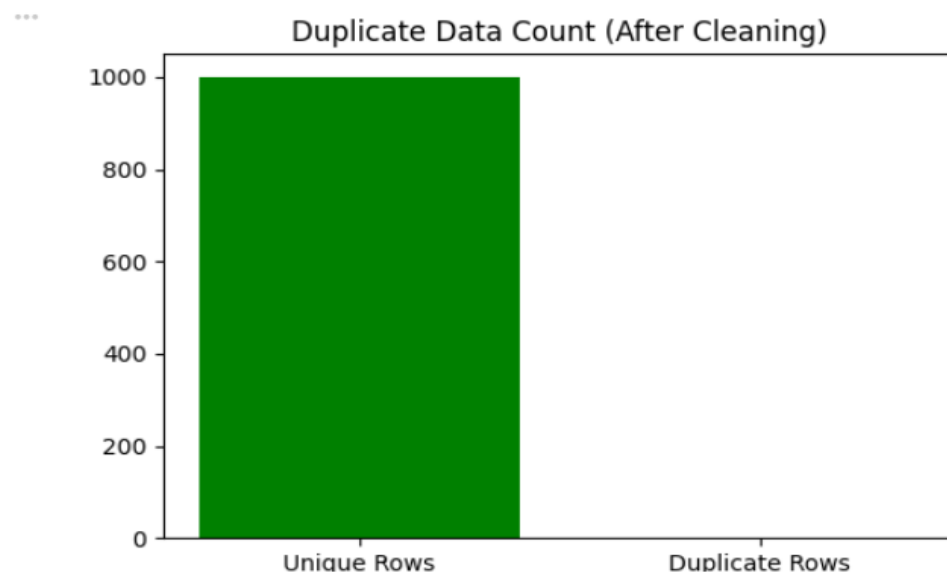
1.2 InfoGainAttributeEval



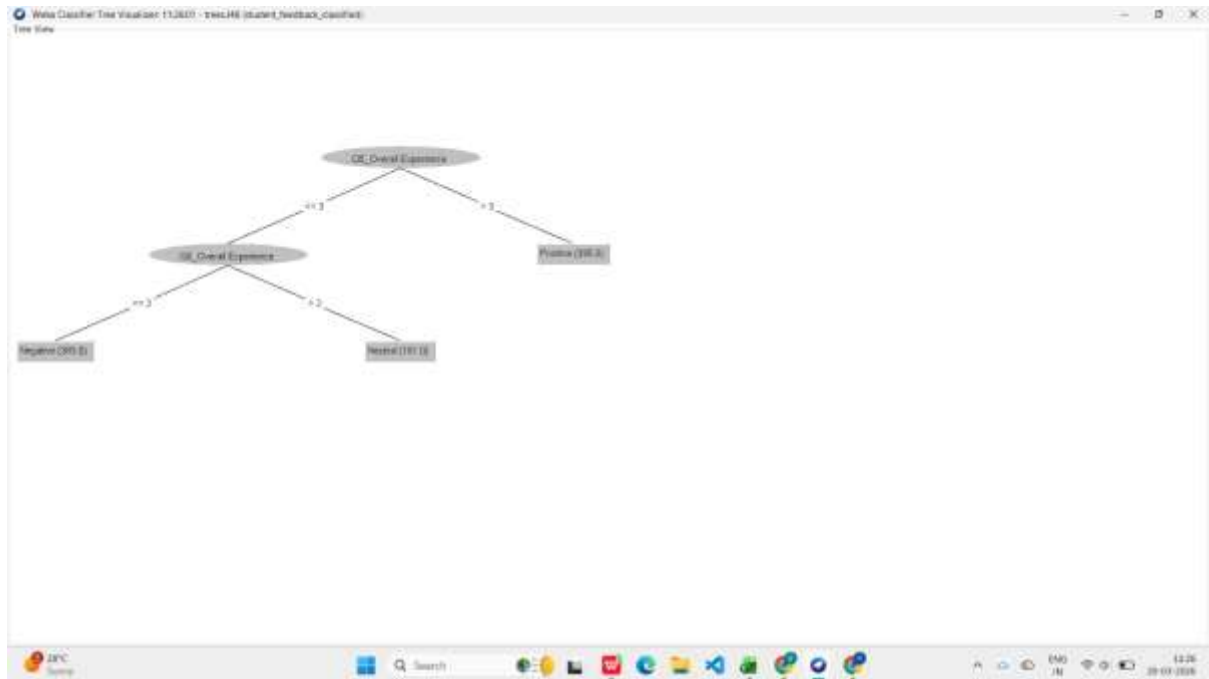
1.3 Classification

```
# After Cleaning
duplicates_after = df_clean[df_clean.duplicated()]
plt.figure(figsize=(6,4))
plt.bar(["Unique Rows", "Duplicate Rows"],
        [df_clean.shape[0]-duplicates_after.shape[0],
         duplicates_after.shape[0]],
        color=["green", "red"])

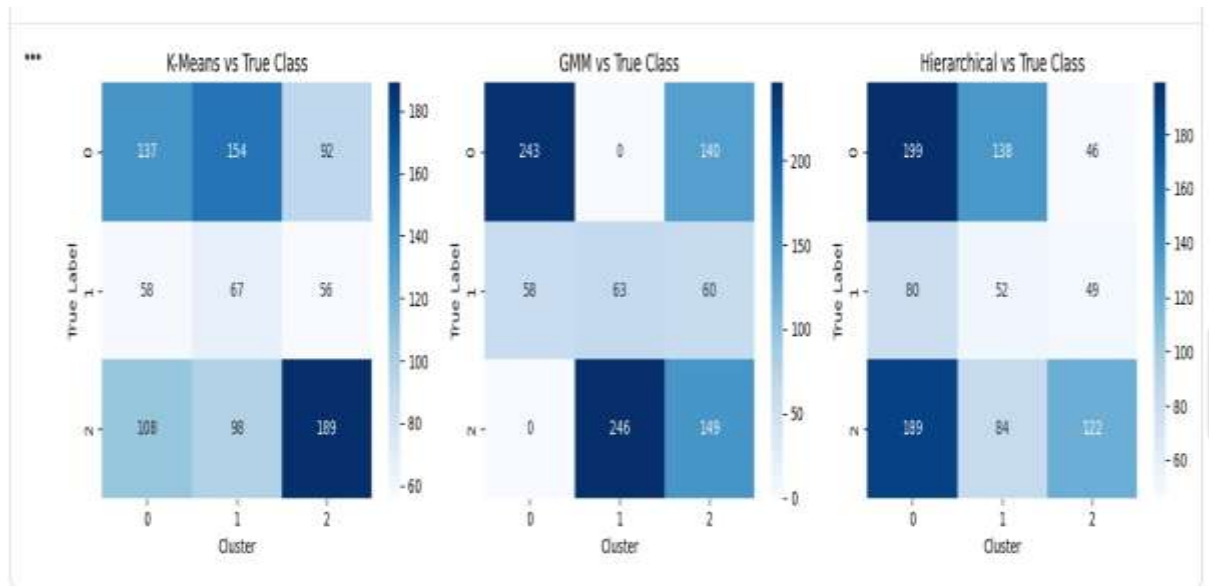
plt.title("Duplicate Data Count (After Cleaning)")
plt.show()
```



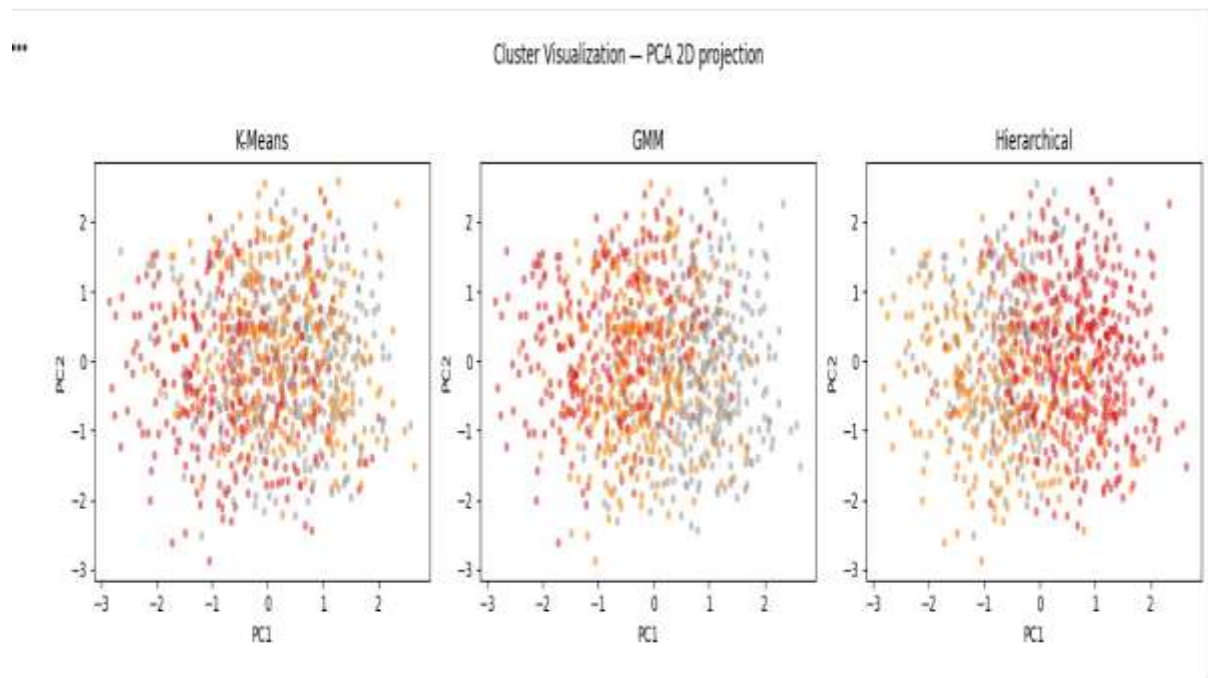
1.4 Data After Cleaning



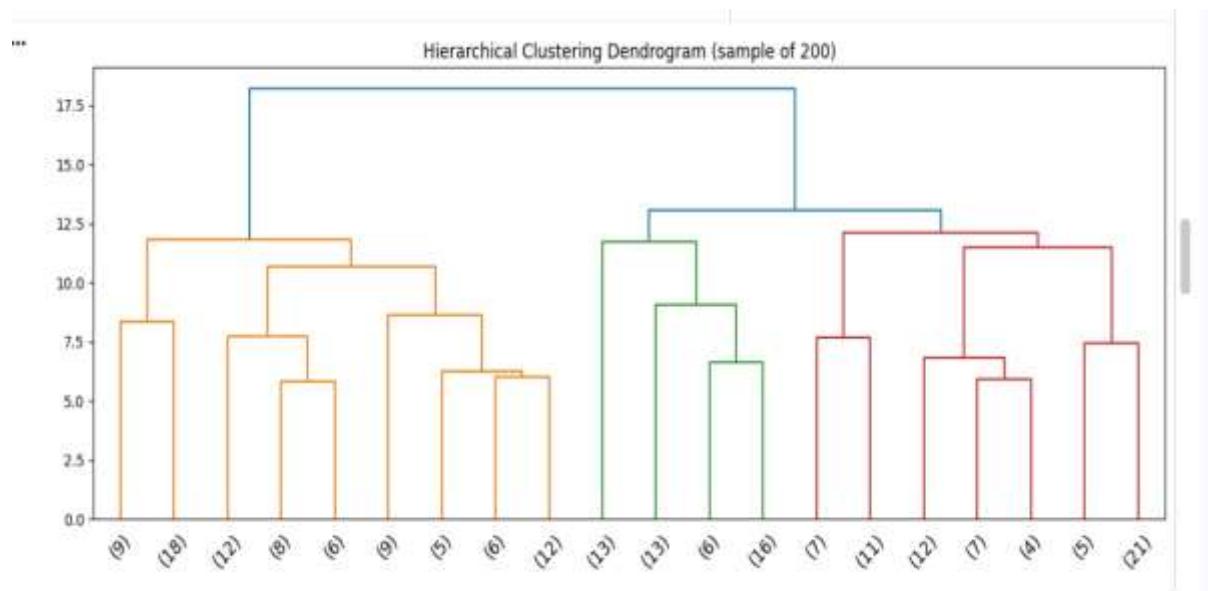
1.5 J48 Decision Tree



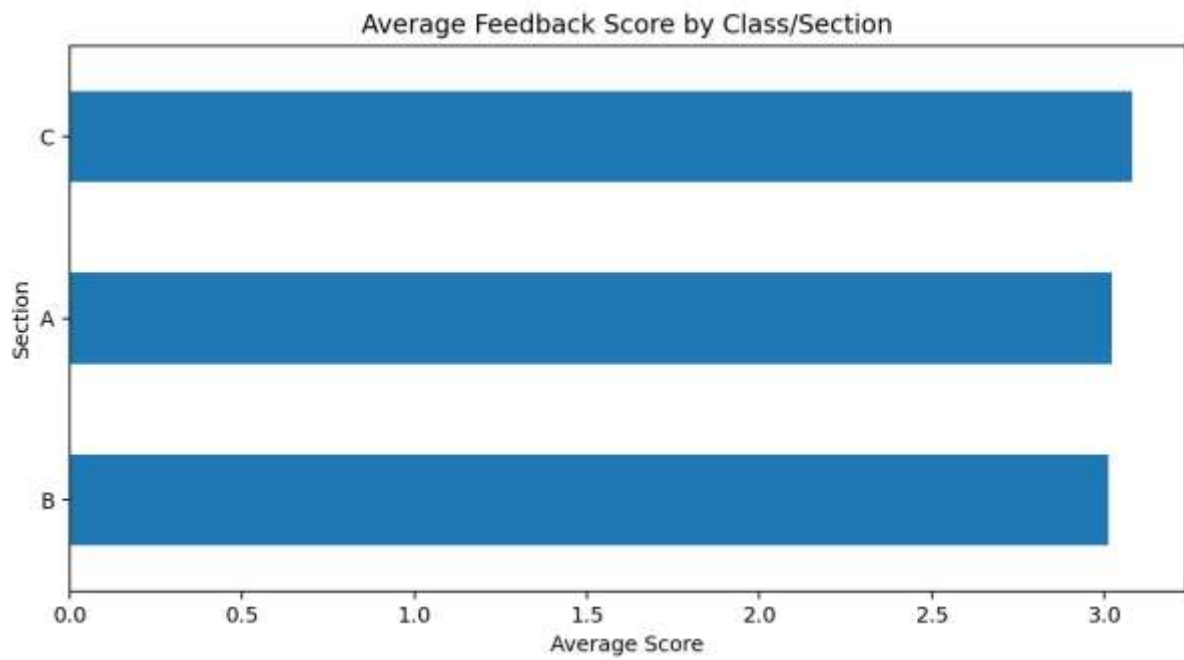
1.6 Different Types of Clusters



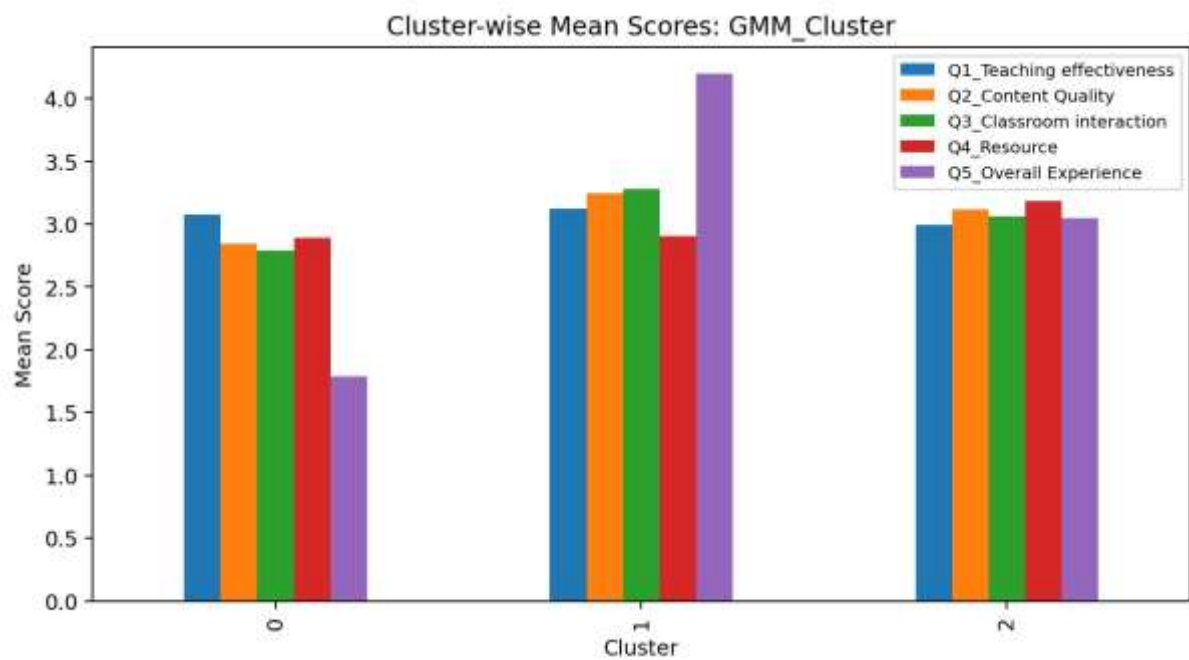
1.7 Cluster Visualizzation



1.8Hirarical Clustering Dendogram



1.9 Average Feedback Score by Class/Section



1.10 Cluster wise Mean Score: GMM_Cluster